

Celebal Excellence Internship

[Week 7 Task – Delta Lake]

Title: CEI – week 7 task submission

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Objective

The objective of this assignment was to perform data exploration, data cleaning, and incremental data processing using Pandas, Apache Spark, and Delta Lake on the Databricks platform.

Work Performed

1. Data Loading

- Loaded the Superstore dataset into a Pandas DataFrame.
- Verified that the dataset was imported successfully.

2. Data Exploration

- Explored the dataset using:
 - head()
 - shape
 - columns
 - data types
- Reviewed the overall structure of the dataset.

3. Data Cleaning

- Checked for missing values.
- Removed duplicate records.
- Filtered data based on business requirements.
- Selected required columns.
- Created derived columns:
 - Unit_Price
 - Total_Amount

- Saved the cleaned dataset as a CSV file.

4. Spark DataFrame Conversion

- Converted the cleaned Pandas DataFrame into a Spark DataFrame.
- Verified the schema and record count.

5. Delta Lake Implementation

- Created a Delta table from the Spark DataFrame.
- Standardized column names to meet Delta Lake requirements.
- Created an incremental dataset to simulate new incoming records.

6. MERGE Operation

- Performed the Delta Lake MERGE operation.
- Updated existing records based on the matching Row_ID.
- Inserted new records that were not present in the master table.

7. Validation

- Verified that the MERGE operation completed successfully.
- Validated the updated records.
- Verified newly inserted records.
- Checked the final row count of the Delta table.

Technologies Used

- Python
- Pandas
- Apache Spark (PySpark)

- Delta Lake
- Databricks

Outcome

This assignment demonstrated the complete workflow of loading, cleaning, transforming, and processing data using Pandas and Spark. The implementation of the Delta Lake MERGE operation successfully handled incremental data by updating existing records and inserting new records into the Delta table. The final solution validated the correctness of the MERGE process and showcased practical data engineering concepts used in modern ETL pipelines.